

microcontroller with Wi-Fi, a Lithium-polymer battery, a built-in radio and the required application programming interface (API). The board is Arduino-compatible, so you can use the Arduino IDE too for programming it.

Hackable hardware

Trend: Truly, everybody can become an inventor with open hardware like Arduino, Raspberry Pi, Beagleboard and now Intel's MinnowBoard! They are small and handy, have trendy features and are easy on the pocket (except perhaps the MinnowBoard, which appears to be costlier than its counterparts). As a result, they are creating a revolution in hands-on learning and everyday innovation. Open hardware, which fosters a hacking culture, is used increasingly for STEM (Science, Technology, Engineering, Mathematics) learning in schools and colleges. It is being used by students, hobby groups and even scientists to develop futuristic devices—socially beneficial ones as well as fun projects. From devices that can feed cats, water plants and guard the home to those that can monitor pollution and improve food production, a lot of amazing stuff is being made with today's 'hackable hardware'.

Thiyagu Palanisamy, head of technology, ThoughtWorks India, says, "Open source hardware is one of the most notable emerging technologies. Hardware design details including schematics, bill of materials (BOM) information, and printed circuit board layout details are made publicly available. This lowers the barrier to entry tremendously, and gives people the freedom to control their technology at the lowest level. Open source hardware doesn't mean free hardware. One would still need to invest in the hardware, but the basic principle is to go with widely-available components. With a little programming experience, anyone can use low-cost computer boards like Arduino, BeagleBone or Raspberry Pi to power any electronic device, from video game pads to unmanned aerial vehicles. This can make a big impact in developing countries like India, where we need low-cost solutions to make technology available for all. Open source hardware will act as a catalyst for simple innovations like building low-cost Wikipedia servers for rural schools."

Trend+: Arduino, Raspberry Pi, Texas Instruments' BeagleBoard (or its barebones version that is more popular today—the BeagleBone), Intel's MinnowBoard, and their variants, are becoming quite popular today. The teams are rapidly adding more and more capabilities to these small single-board microcontrollers, and people are harnessing their power at an even faster rate to build a lot of stuff including satellites!

ArduSat, an open source, CubeSat-based nano-satellite built using Arduino boards and sensors is an interesting example. While the satellite is in space, common people can use this set-up for their own experiments too. The Indian organisation Dhruva Space has tied up with Nanosat5

LLC, the makers of ArduSat, to bring its benefits to India. Dhruva Space welcomes people to sign-up for a three-day or one-week access to ArduSat to learn more about how they can use it for their benefit.

Another interesting Raspberry Pi-based project is BrickPi, an open source slide-on board for the Raspberry Pi that helps you connect LEGO blocks, sensors, motors and parts to easily turn the compact Raspberry Pi into a powerful robot. You can connect up to three motors and up to four sensors, both digital and analogue. This is a wonderful way to help kids delve into the wonders of electronics and robotics.

Actually, the number of projects based on these open source boards is just mind-boggling, and a Web search could send you into device wonderland!

Defending the borders

Trend: The defence space is where the cutting edge is. Here, lives are at stake. Open source has found a place in this vertical, even though Oracle may disagree (<http://www.oracle.com/us/products/middleware/cloud-app-foundation/weblogic/dod-and-open-source-software-2012277.pdf>). Apart from using open source platforms for military intelligence systems, the defence departments of several countries are also using open source design tools to make their equipment. Recently, they have also begun to experiment with crowdsourcing to create military equipment designs.

Trend+: Technologies like cloud computing and open designs (of everything from guns to tanks) have definitely changed the performance of the military, irreversibly. One example is the Distributed Common Ground System-Army Standard Cloud (DSC) deployed in Afghanistan, a solution that gives soldiers full access to intelligence and enables fast responses to queries. The DSC has been built using open source platforms and software. The network consists of 228 servers, 1800+ CPUs, 100 dual six-core processors operating at 2.80 GHz and 128 dual quad-core processors running at 2.93 GHz. It requires 13.92 terabytes of RAM and 1.032 TB of hard drive storage. The five key technologies at work in the DSC are Hadoop, Hadoop Core, Accumulo, Condor and SolrCloud.

Recently, the US army also collaborated with crowdsourcing experts to launch ArmyCoCreate.com, to enable soldiers, designers and engineers collaborate on identifying soldier requirements and to then design prototypes to address them. Defense Distributed is another super example of a crowd-funded online open source organisation that designs open source military weapons that can be 3D printed.

FOSS for society

Trend: Open source-based projects with a social slant are on the rise. And this is perhaps one of the most beneficial amongst the emerging trends. There are community projects that aim to monitor and control pollution, track garbage disposal, help people in distress, prevent eve-teasing, ensure drought or